CSE 312

Foundations of Computing II

Lecture 1: Introduction & Counting

https://cs.washington.edu/312

Instructors

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A Team of fantastic TAs



Adin Aberbach



Beta Chen



Darin Ershov



Arnav Mazumder



Medha Mittal



Gaurang Pendharkar



Jack Scott



Sam Sullivan



Mayee Sun



Glenn Sun



Toby Thornburg

See https://courses.cs.washington.edu/courses/cse312/26au/staff.php to learn more about the staff and office hours.

Lectures and Sections

- **Lectures MWF [CSE2 G01 9:30-10:20, MGH 389 1:30-2:20]**
- Classes will be in person
- Lecture recording but no live streaming because of classroom

- **Poll Everywhere**
  - We will sometimes use Poll Everywhere during class
  - You sign up directly

Go to https://www.polleverywhere.com/login and login using YOURNETID@uw.edu

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- **Sections Thu (starts this week)**
  - Not recorded
  - Will prepare you for problem sets!

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- **Ed Discussion**
  - You should have received an invitation (synchronized with the class roaster)
  - Material (resources tab)
  - Announcements (discussion tab)
  - Discussion (discussion tab)

Use Ed discussion forum as much as possible. You can make private posts that only the staff can view! Email instructor for personal issues.

Engagement

- **"Concept checks" after each lecture 3%**
  - Due on Gradescope at 1:00 pm on the day of the next lecture.
  - Simple questions to reinforce concepts taught in each class
  - Keep you engaged throughout the week, so that homework becomes less of a hurdle
- **9 Problem Sets (Gradescope) 25%**
  - Solved individually. Discussion allowed; list collaborators and write your own solutions.
  - Generally due Wednesdays until Thanksgiving, Fridays after that.
  - Check the course website for the first problem set release.
- **Midterm 24%**
  - In class on **Wednesday, Nov 4**
- **Final Exam 48%**
  - **Monday, December 14, 12:30-2:20 pm: Section A in BAG 131; Section B in BAG 154.**

See the syllabus for late-day and concept-check policies.

For more details see

Course Webpage https://cs.washington.edu/312

Foundations of Computing II

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Introduction to Counting, Probability & Statistics

for computer scientists

What is probability??

Why probability?!

Probability is our tool for understanding random processes

- Randomness in nature and the sciences/social sciences
  - At the quantum level, everything is random
  - Best way to understand and simulate behavior of complex systems
  - A way to design and understand experiments, observations
    - In the lab, the field, medical trials, surveys
- In Computer Science, randomness has these kinds of roles but also important new ones…

Probability and randomness in Computer Science

- Understanding/modelling the inputs to and behavior of our algorithms
  - In ML, program testing/fuzzing, algorithm analysis, …

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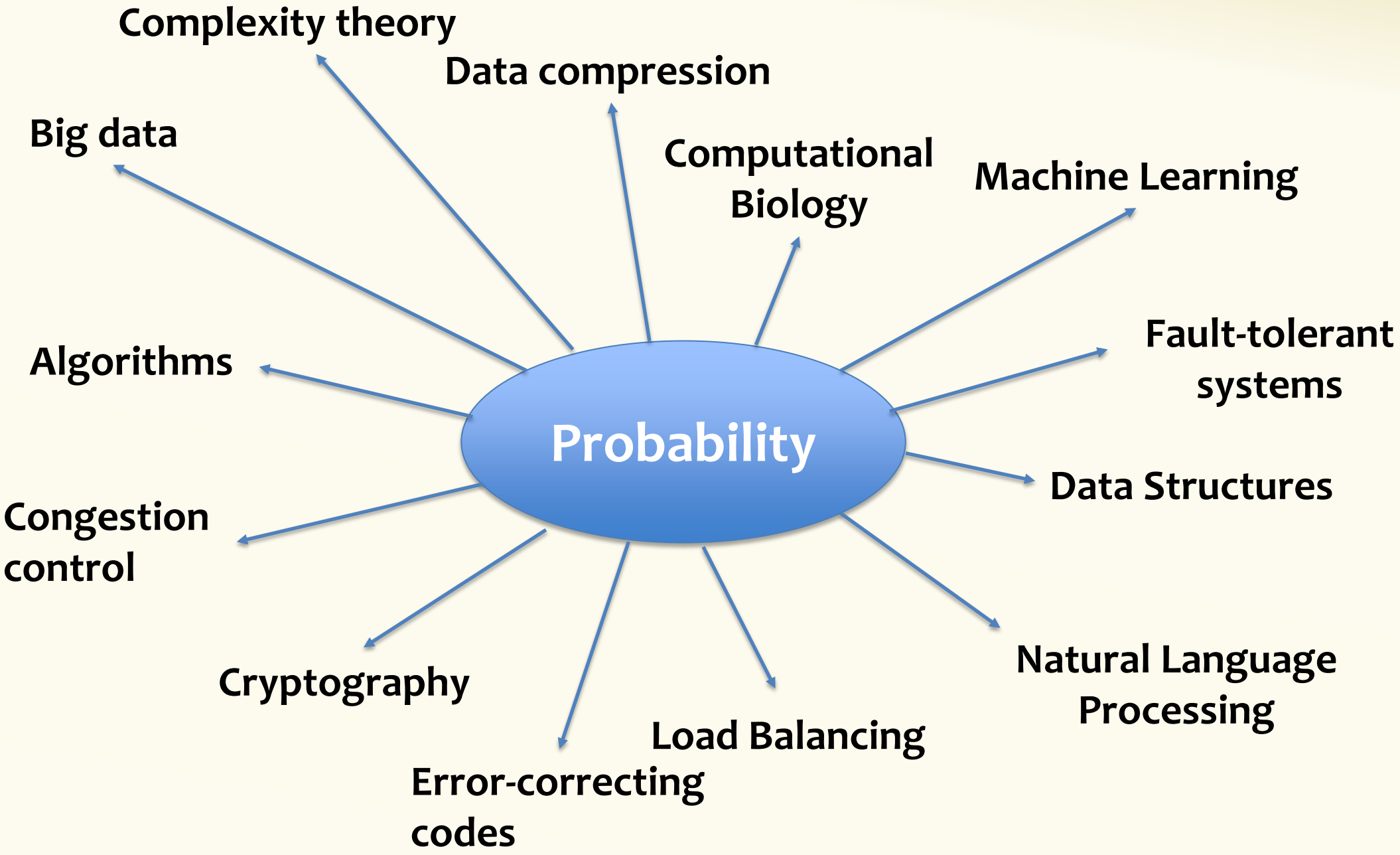
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  - Cryptography, privacy, fault tolerance, computer security, …
- A tool for simpler and more efficient design
  - Data structures, algorithms, ML, …
- …

+ much more!


Complexity theory
Data compression
Big data
Computational Biology
Machine Learning
Algorithms
Fault-tolerant systems
Probability
Data Structures
Congestion control
Cryptography
Natural Language Processing
Load Balancing
Error-correcting codes

Content

- Counting (basis of discrete probability)
  - Counting, Permutation, Combination, inclusion-exclusion, Pigeonhole Principle
- What is probability
  - Probability space, events, basic properties of probabilities, conditional probability, independence, expectation, variance
- Properties of probability
  - Various inequalities, Zoo of discrete random variables, Concentration, Tail bounds
- Continuous Probability
  - Probability Density Functions, Cumulative Density Functions, Uniform, Exponential, Normal distributions, Central Limit Theorem, Estimation
- Applications
  - A sample of randomized algorithms, differential privacy, learning …

Today: A fast introduction to counting so you will have enough to work on in section tomorrow…



We are interested in counting the number of objects with a certain given property.

"How many ways are there to assign 7 TAs to 5 sections, such that each section is assigned to two TAs, and no TA is assigned to more than two sections?"

"How many positive integer solutions (x, y, z) does the equation $x^3 + y^3 = z^3$ have?"

Generally: Question boils down to computing cardinality $|S|$ of some given set S.

(Discrete) Probability and Counting are Twin Brothers

"What is the probability that a random student from CSE312 has black hair?"

$$= \frac{\#\ students\ with\ black\ hair}{\#students}$$



Today – Two basic rules

- Sum rule
- Product rule

Sum Rule

If you can choose from

- EITHER one of n options,
- OR one of m options with NO overlap with the previous n

then the number of possible outcomes of the experiment is

$$n + m$$

Counting “lunches”

If a lunch order consists of **either** one of 6 soups **or** one of 9 salads, how many different lunch orders are possible?

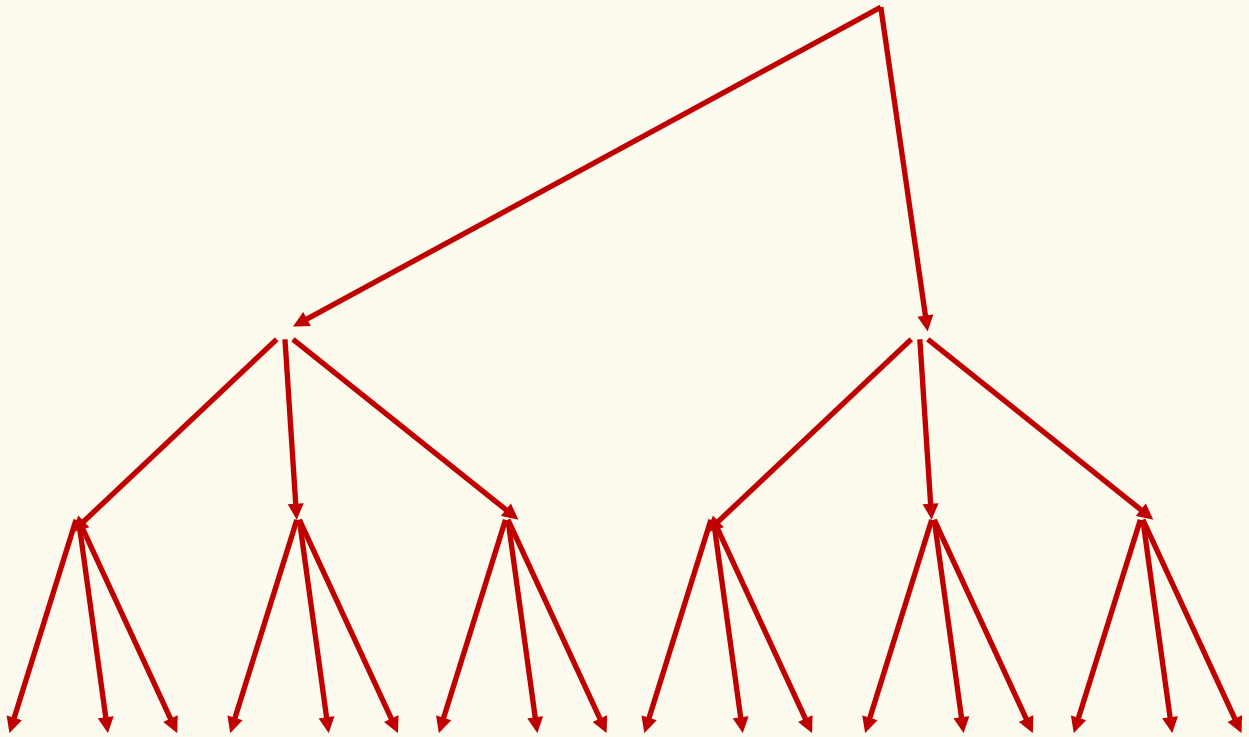




Product Rule: In a sequential process, there are

- n_1 choices for the first step,
- n_2 choices for the second step (given the first choice), …, and
- n_m choices for the mth step (given the previous choices),

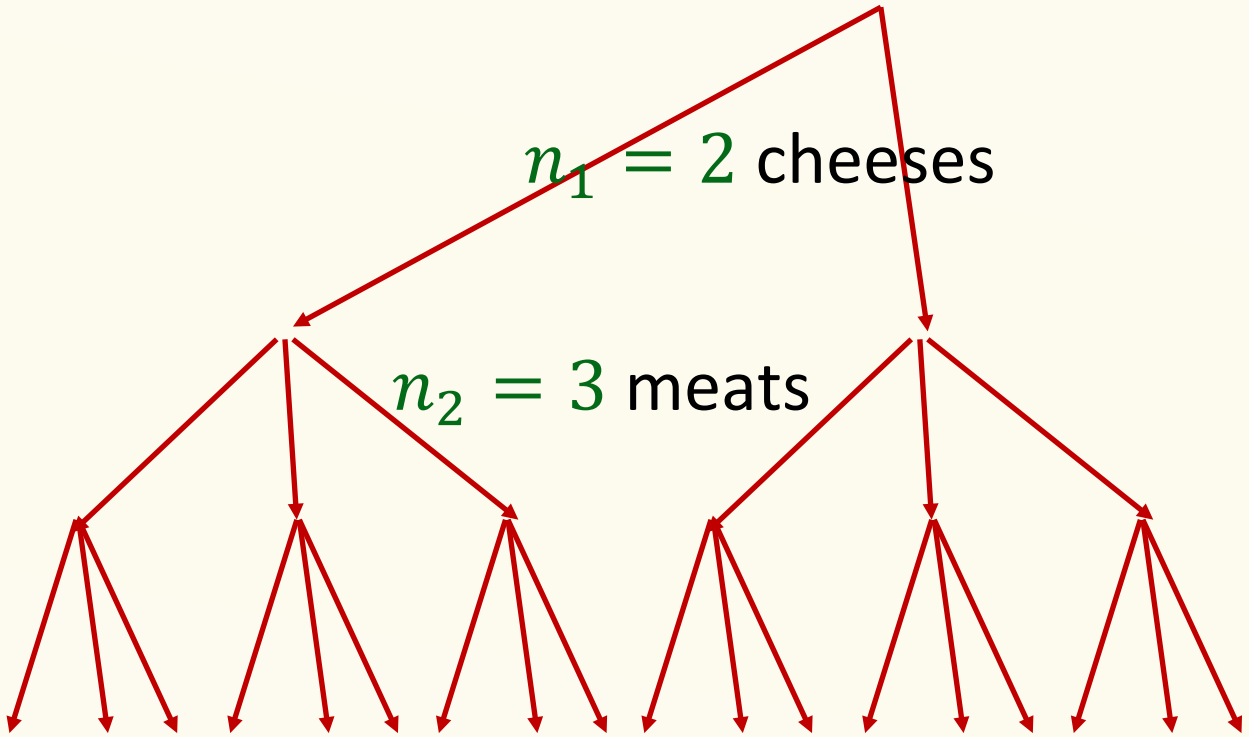
then the total number of outcomes is $n_1 \times n_2 \times \cdots \times n_m$



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Example: "How many subs?"

$\square \times \square \times \square = \square$

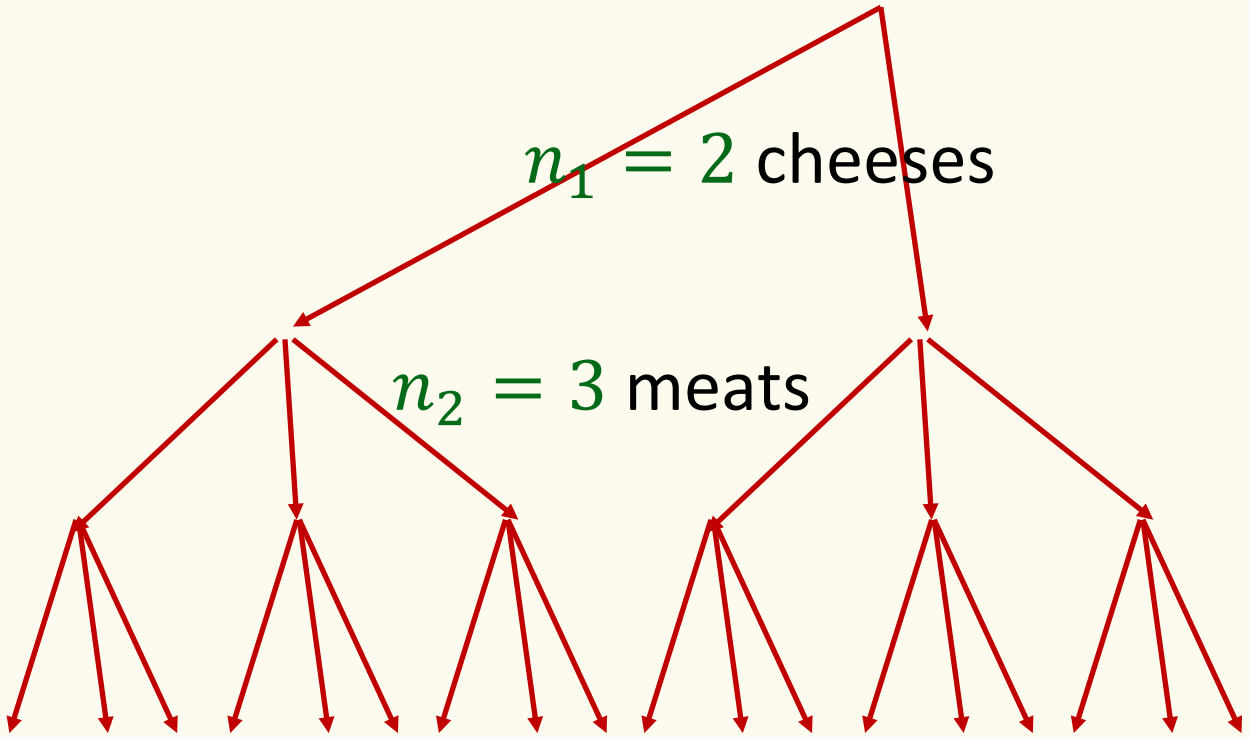
$n_3 = 3$ veggies



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Example: "How many subs?"

2	×	3	×	3	=	18

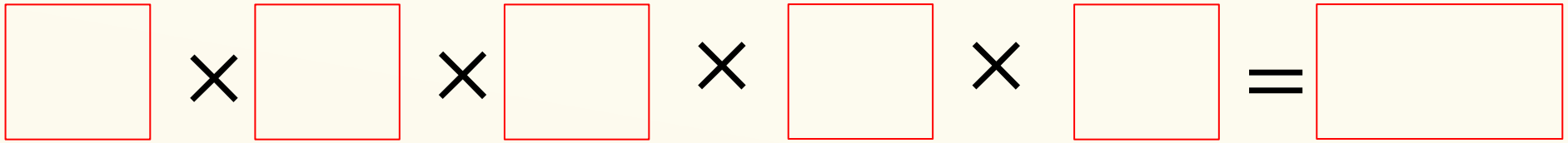
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Product rule examples – Strings

How many strings of length 5 over the alphabet $\{A, B, C, \ldots, Z\}$ are there?

- E.g., AZURE, BINGO, TANGO, STEVE, SARAH, …



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$$\boxed{26} \times \boxed{26} \times \boxed{26} \times \boxed{26} \times \boxed{26} = \boxed{(26)^5}$$

How many binary strings of length n over the alphabet $\{0,1\}$?

- E.g., $0 \cdots 0$, $1 \cdots 1$, $0 \cdots 01$, …

$$\boxed{} \times \boxed{} \times \boxed{} \times \cdots \times \boxed{} = \boxed{}$$

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$$\boxed{2} \times \boxed{2} \times \boxed{2} \times \cdots \times \boxed{2} = \boxed{2^n}$$

Product rule example – Cartesian Product

Definition. The **cartesian product** of two sets S, T is

$$S \times T = \{(a, b) : a \in S, b \in T\}$$

Called a 2-sequence
Order matters! $(a, b) \neq (b, a)$

$$|S \times T| = \square \times \square$$

$$|A_1 \times A_2 \times \cdots \times A_n| = \square \times \square \times \square \times \cdots \times \square$$

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Product rule example – Power set

Definition. The **power set** of S is the set of all subsets of S,

$$\{X : X \subseteq S\}.$$

Notations: $\mathcal{P}(S)$ or simply 2^S (which we will use).

Example. $2^{\{\bigstar,\spadesuit\}} = \{\emptyset, \{\bigstar\}, \{\spadesuit\}, \{\bigstar, \spadesuit\}\}$

$2^{\emptyset} = \{\emptyset\}$

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How many different subsets of S are there if $|S| = n$?

Product rule example – Power set

set $S = \{e_1, e_2, e_3, \cdots, e_n\}$

subset $X = \{$ $\}$

$\square \times \square \times \square \times \cdots \times \square = \square$

Product rule example – Power set

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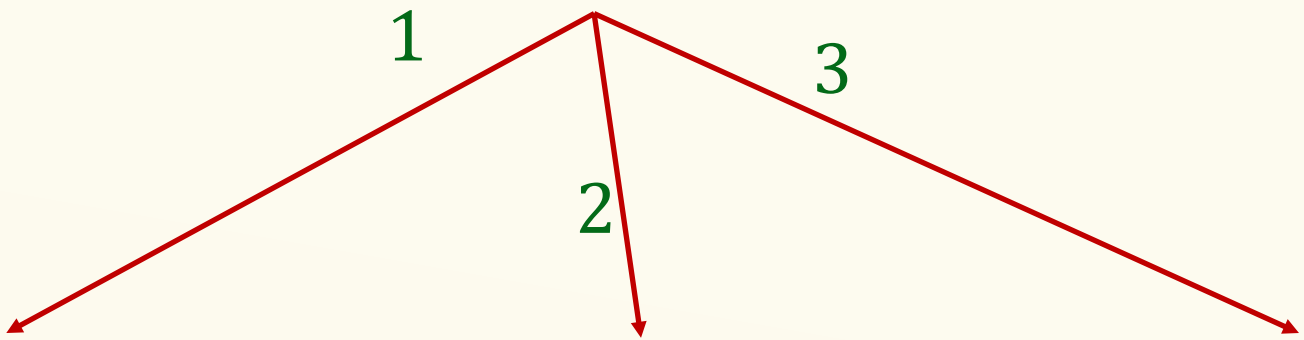
subset $X = \{ \quad \}$

$$\boxed{2} \times \boxed{2} \times \boxed{2} \times \cdots \times \boxed{2} = \boxed{2^{|S|}}$$

Proposition. $|2^S| = 2^{|S|}$

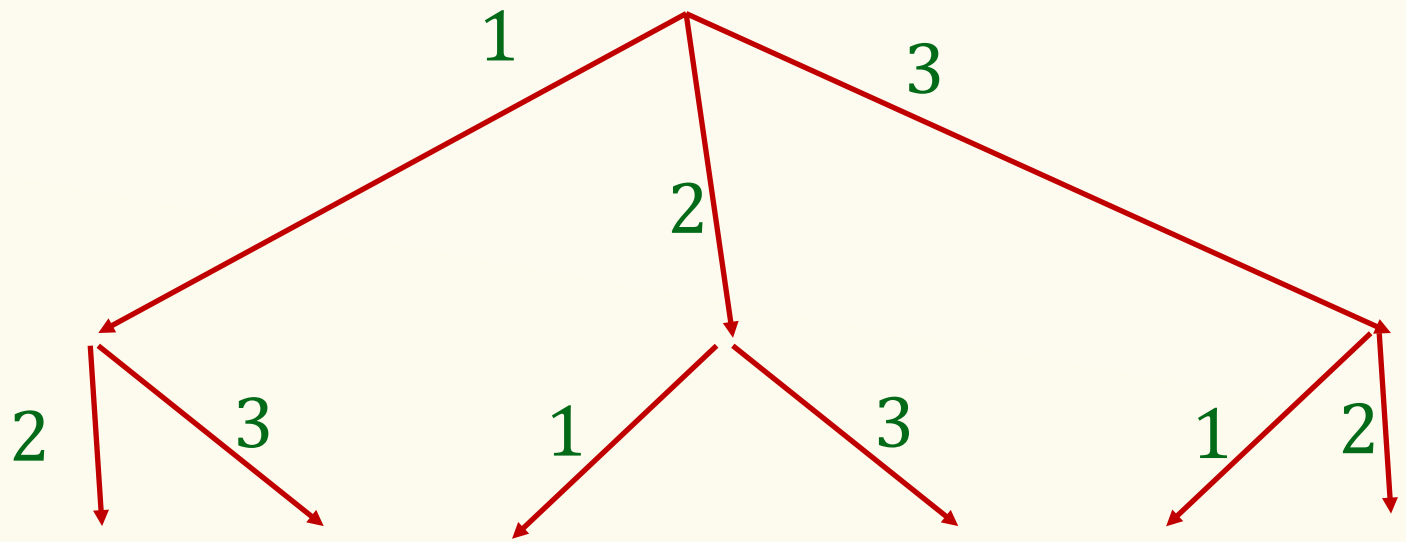
Note: Sequential process for product rule works even if the sets of options are different at each point

"How many sequences in $\{1,2,3\}^3$ *with no repeating elements?"*



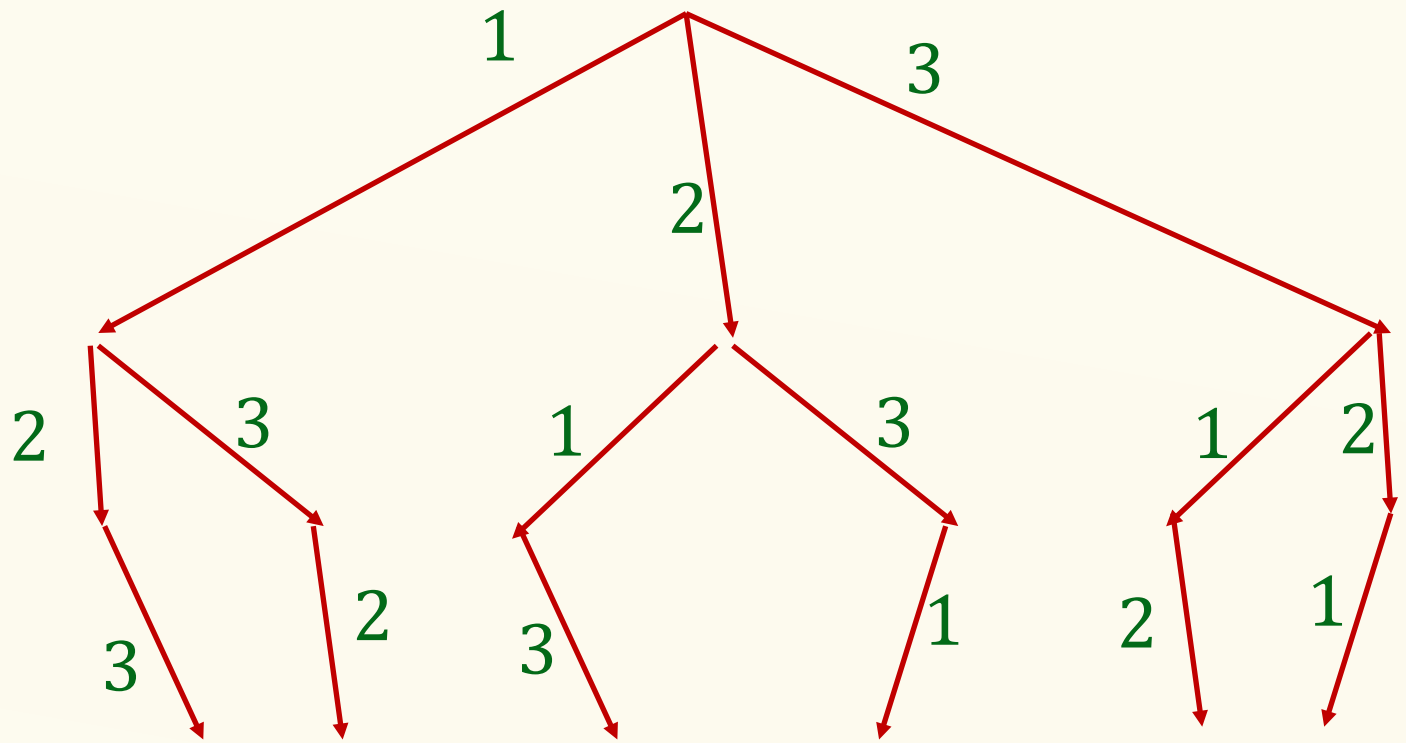
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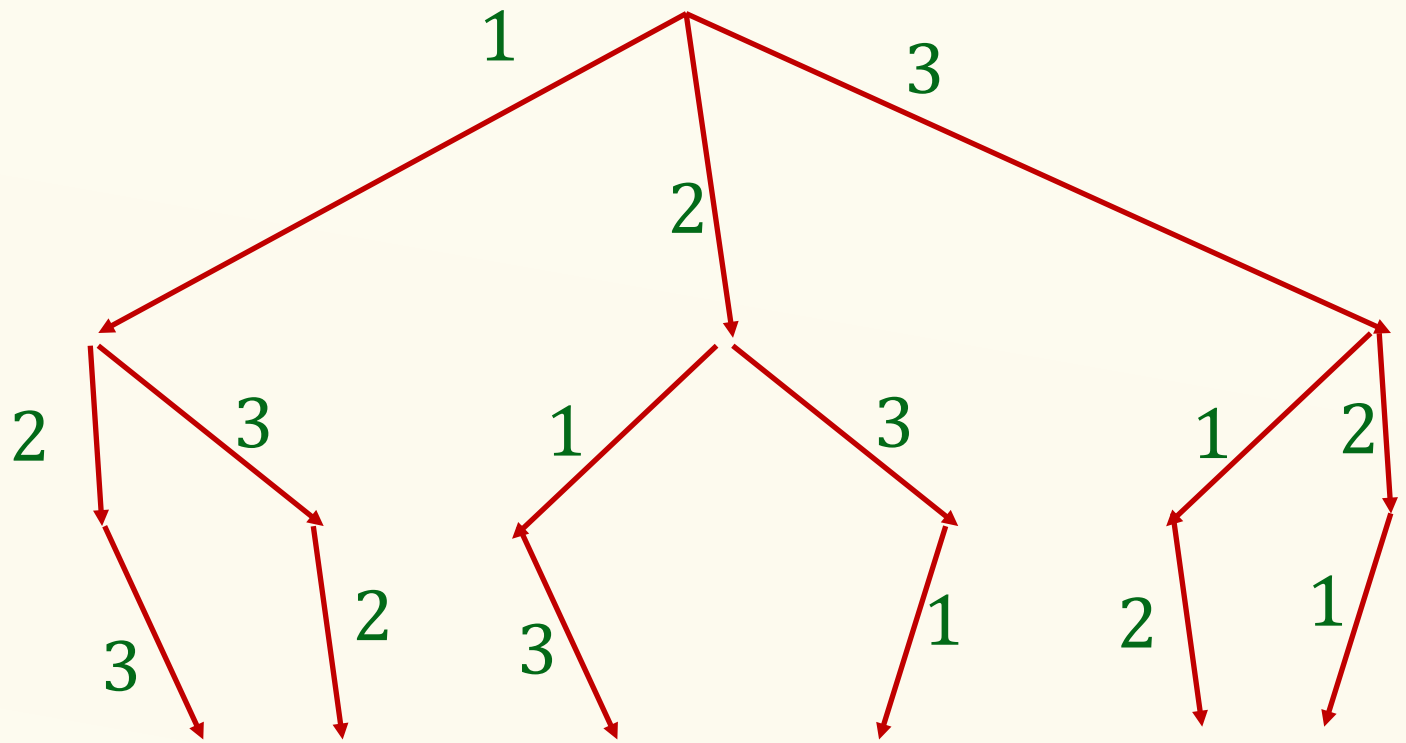
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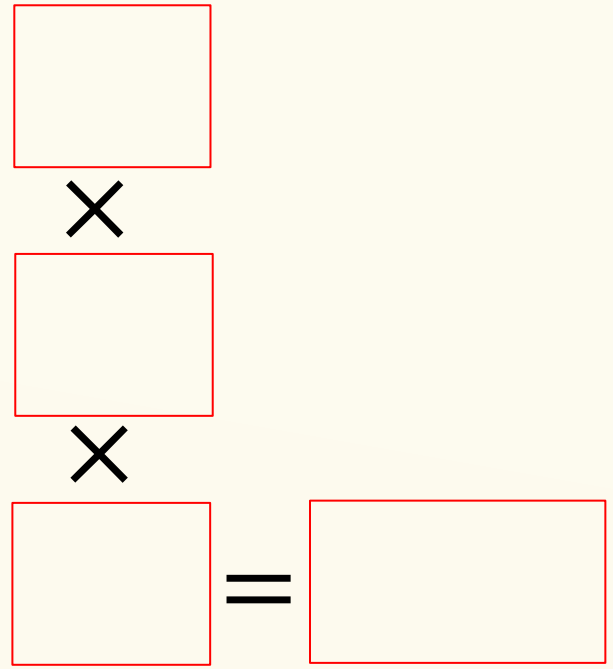
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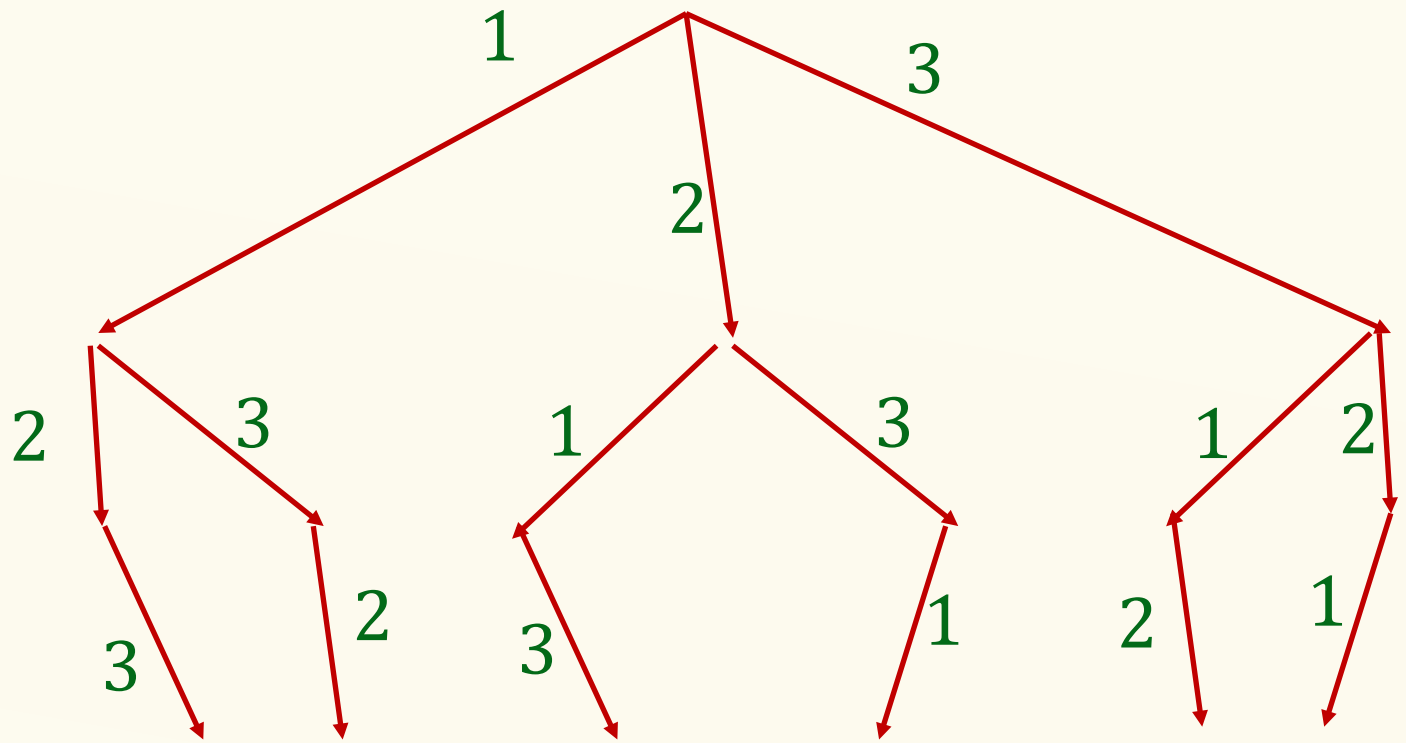
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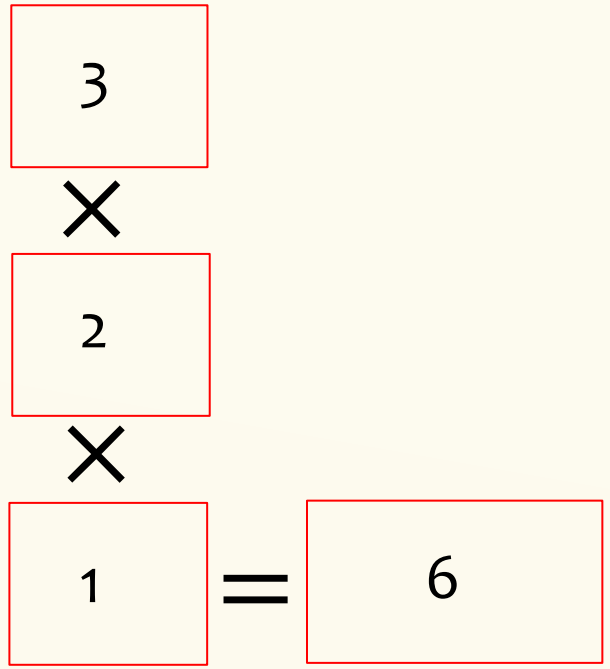




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Factorial

"How many ways to order elements in S, where $|S| = n$?"

"Permutations"

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Answer = $n \times (n-1) \times (n-2) \times \cdots \times 2 \times 1$

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Definition. The **factorial function** is

$$n! = n \times (n-1) \times \cdots \times 2 \times 1$$

Note: $0! = 1$

Nice use of sum rule: Counting using complements

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"# of sequences in $\{1,2,3\}^3$ *with no repeating elements"* $n =$ ☐

"# of sequences in $\{1,2,3\}^3$ $3^3 = 27$

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"# of sequences in $\{1,2,3\}^3$ $3^3 = 27$ $= m + n$ *by the sum rule*



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$m = 27 - n =$ ☐

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"# of sequences in $\{1,2,3\}^3$ *with no repeating elements"* $n =$ 3!=6

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$m = 27 - n =$ 21

Distinct Letters

"How many sequences of 5 ***distinct*** *alphabet letters from* $\{A, B, \ldots, Z\}$*?"*

E.g., AZURE, BINGO, TANGO. But not: STEVE, SARAH

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Answer: $26 \times 25 \times 24 \times 23 \times 22$
$= 7893600$

In general

Fact. # of k-element sequences of distinct symbols from an n-element set is

$$P(n,k) = n \times (n-1) \times \cdots \times (n-k+1) = \frac{n!}{(n-k)!}$$

In general

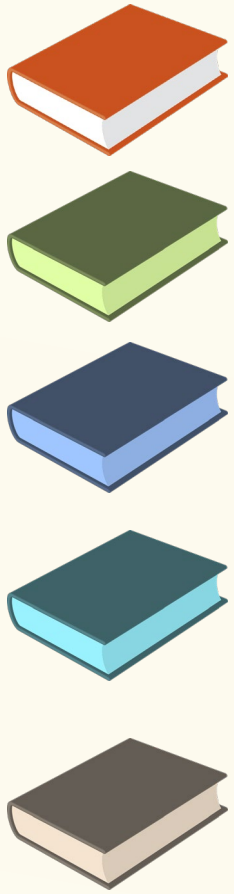
Aka: k-permutations

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Product rule – One more example

5 books



"How many ways are there to distribute 5 books among Alice, Bob, and Charlie?"

Every book to one person, everyone gets ≥ 0 books.



Alice

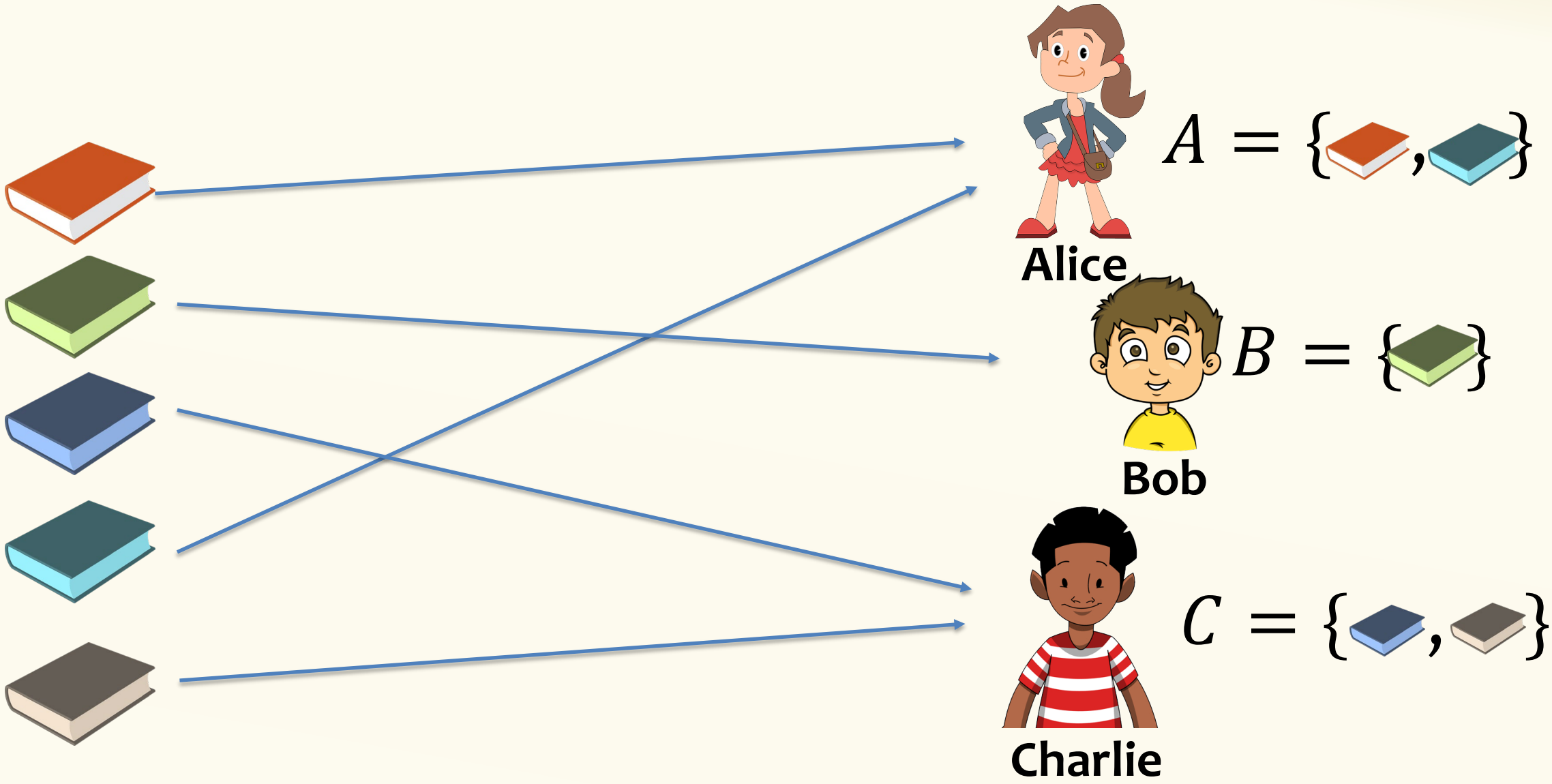


Bob



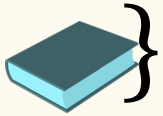
Charlie

Example Book Assignment

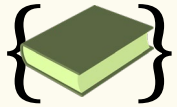


Book assignment – Modeling

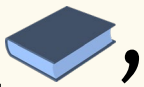
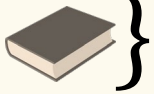


$A = \{$  ,  $\}$



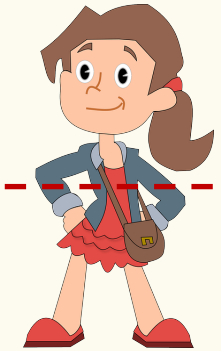
$B = \{$  $\}$

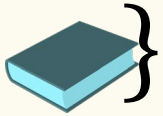


$C = \{$  ,  $\}$

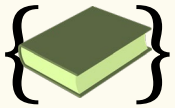
Book assignment – Modeling

$2^5 = 32$ options

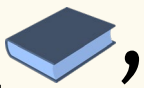


$A = \{$  ,  $\}$



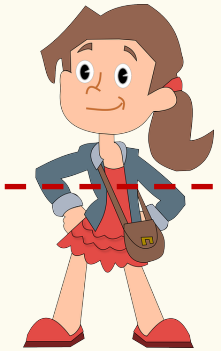
$B = \{$  $\}$

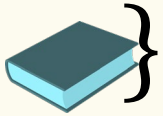


$C = \{$  ,  $\}$

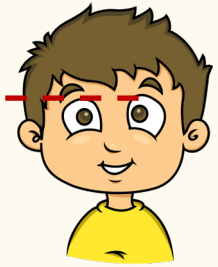
Book assignment – Modeling

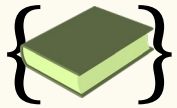
$2^5 = 32$ options



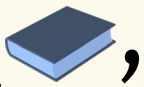
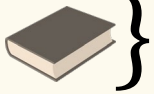
$A = \{$  ,  $\}$

$2^5 = 32$ options



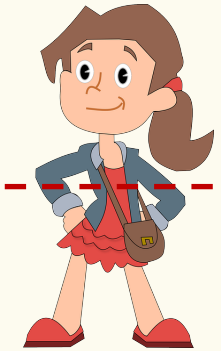
$B = \{$  $\}$

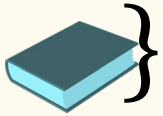


$C = \{$  ,  $\}$

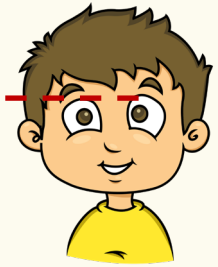
Book assignment – Modeling

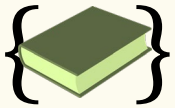
$2^5 = 32$ options



$A = \{$  ,  $\}$

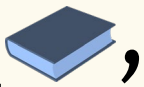
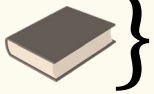
$2^5 = 32$ options



$B = \{$  $\}$

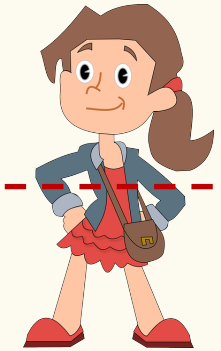
$2^5 = 32$ options

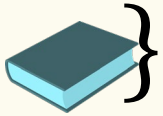


$C = \{$  ,  $\}$

Book assignment – Modeling

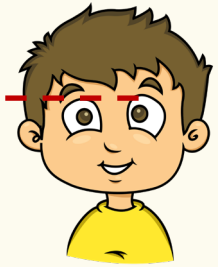
$2^5 = 32$ options

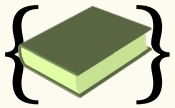


$A = \{$  ,  $\}$

$\times$

$2^5 = 32$ options

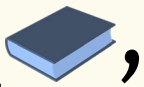
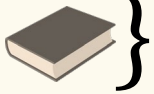


$B = \{$  $\}$

$\times$

$2^5 = 32$ options

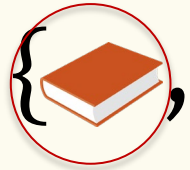
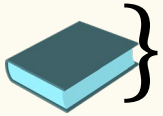


$C = \{$  ,  $\}$

$= 32^3$ assignments

Problem – Overcounting

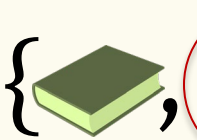


$A = \{$  $,$  $\}$

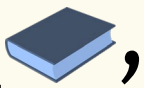
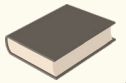
Problem: We are counting some invalid assignments!!!

→ **overcounting!**



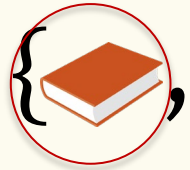
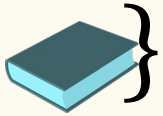
$B = \{$  $,$  $\}$



$C = \{$  $,$  $\}$

Problem – Overcounting

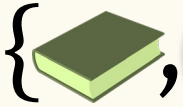


$A = \{$  $,$  $\}$

Problem: We are counting some invalid assignments!!!

→ **overcounting!**



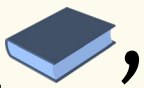
$B = \{$  $,$  $\}$

What went wrong in the sequential process?

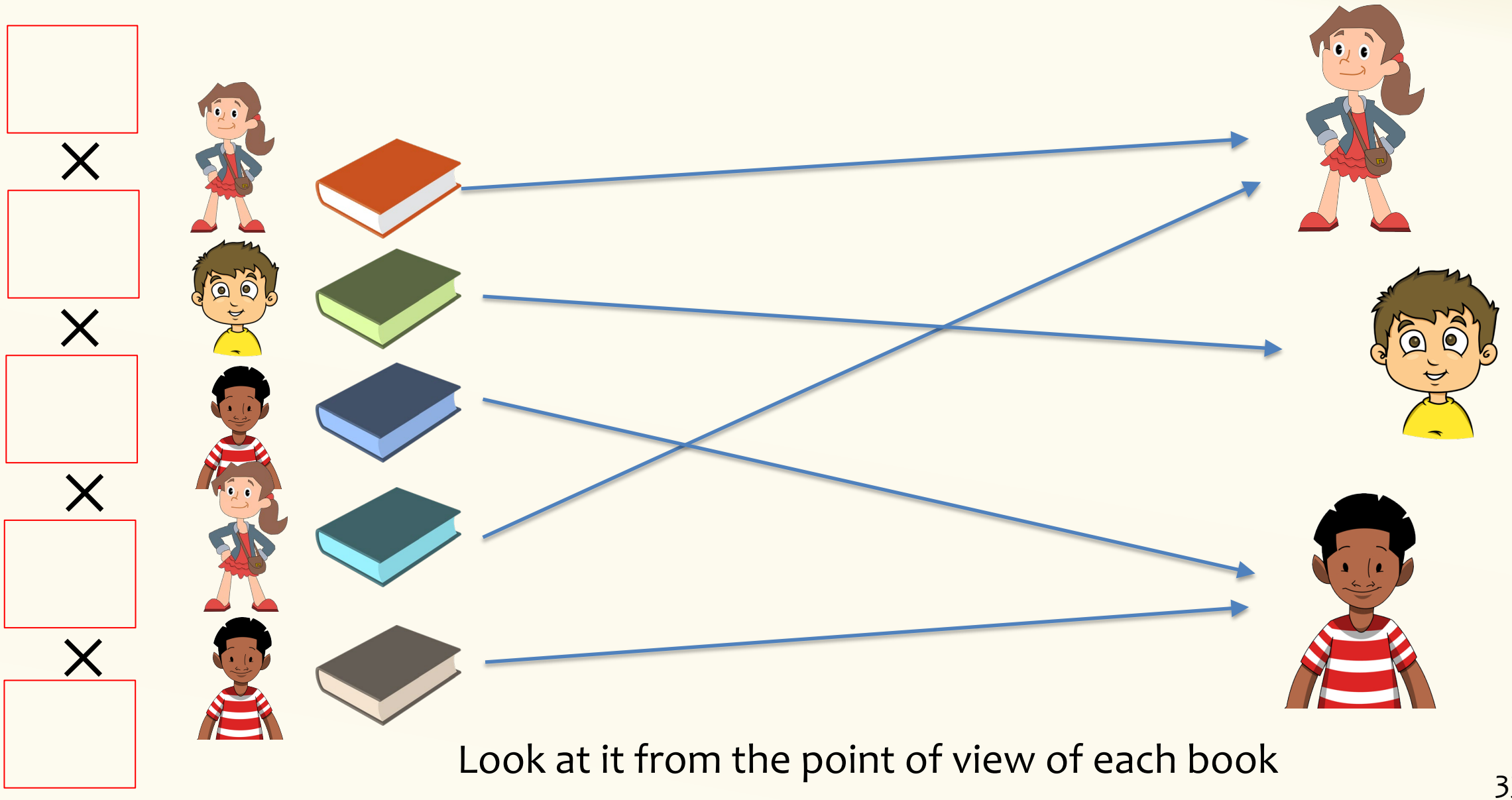
After assigning A to Alice,

B is no longer a valid option for Bob



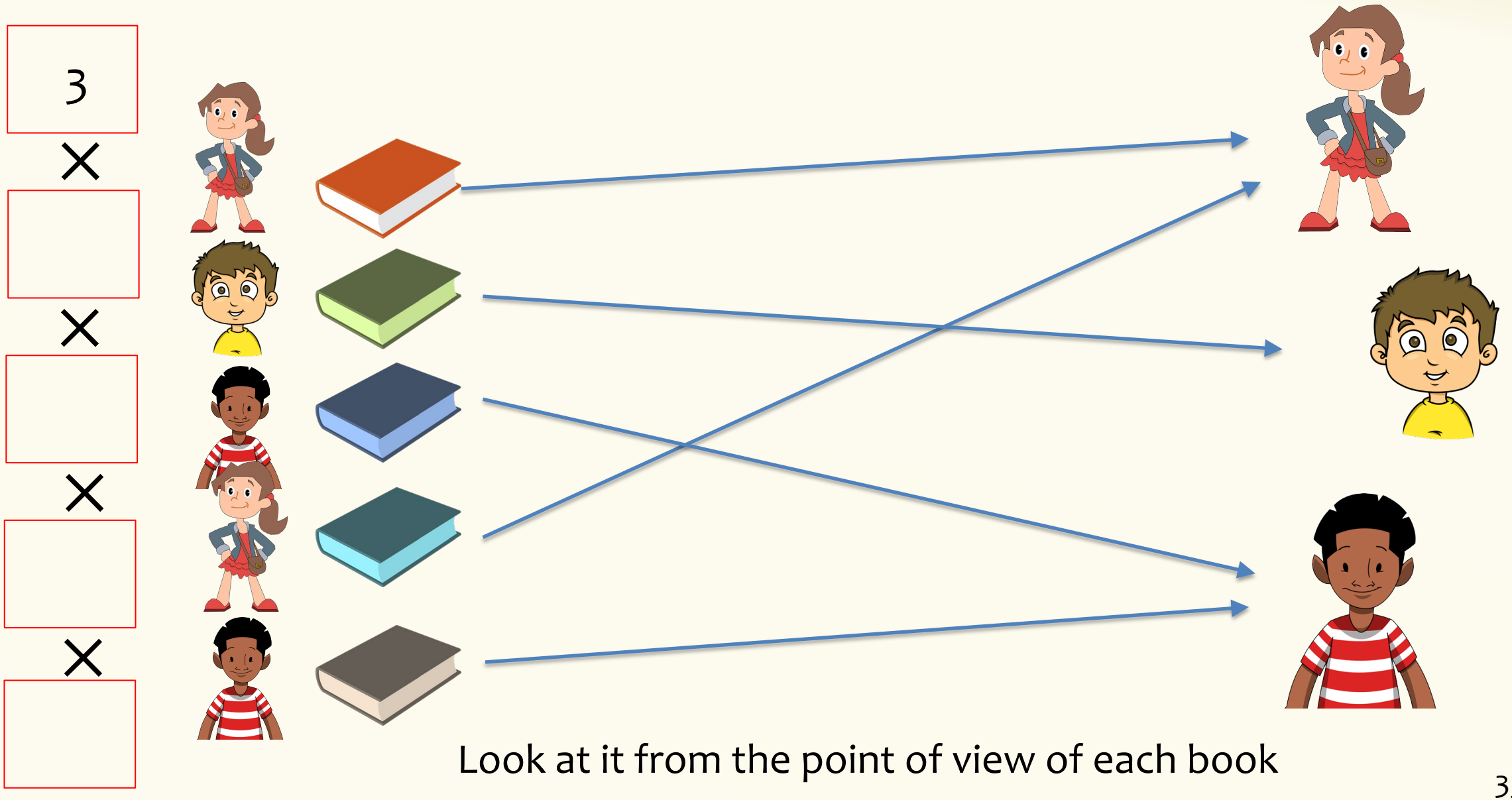
$C = \{$  $,$  $\}$

Book assignments – A Clever Approach



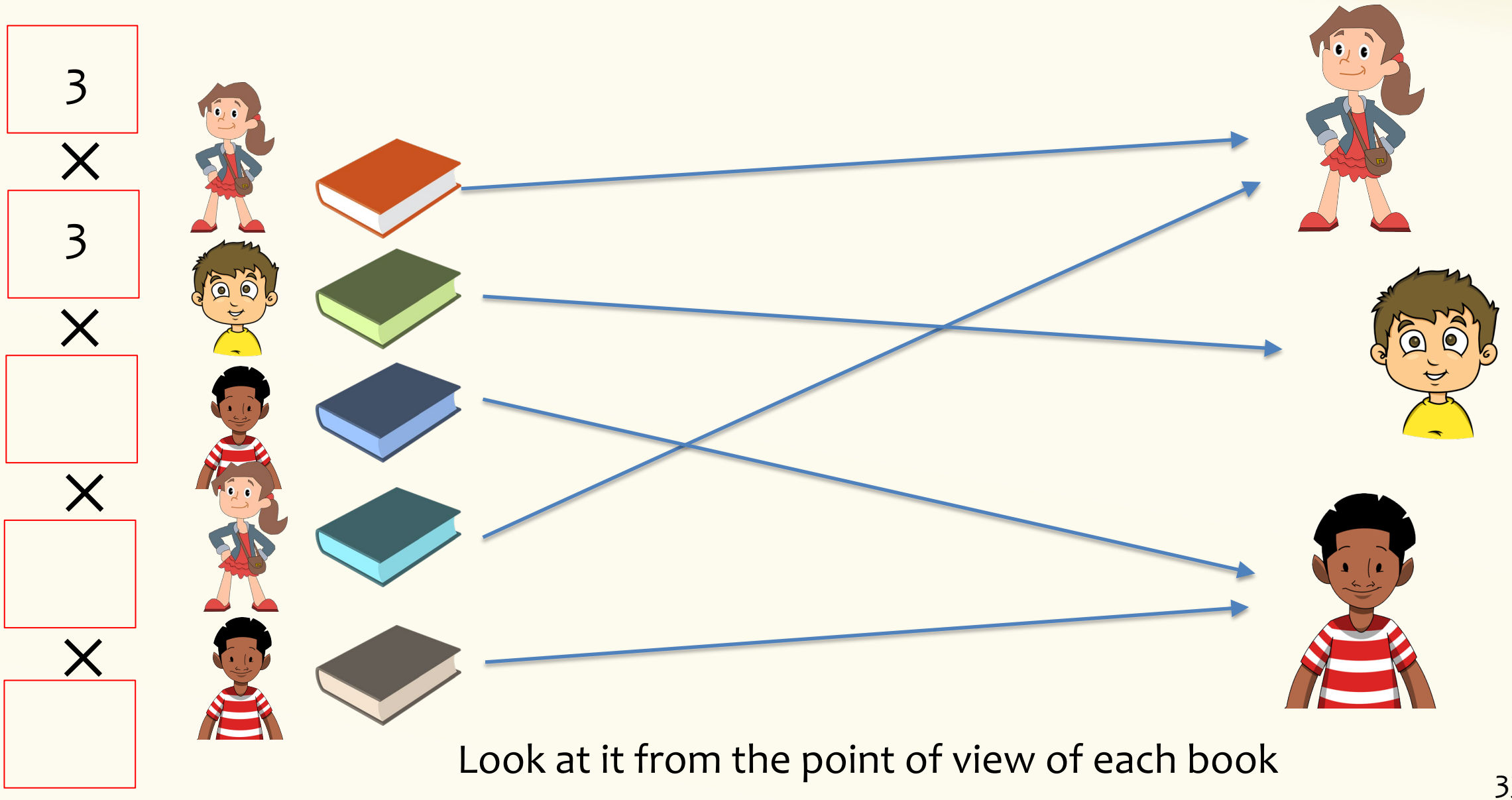
Look at it from the point of view of each book

Book assignments – A Clever Approach



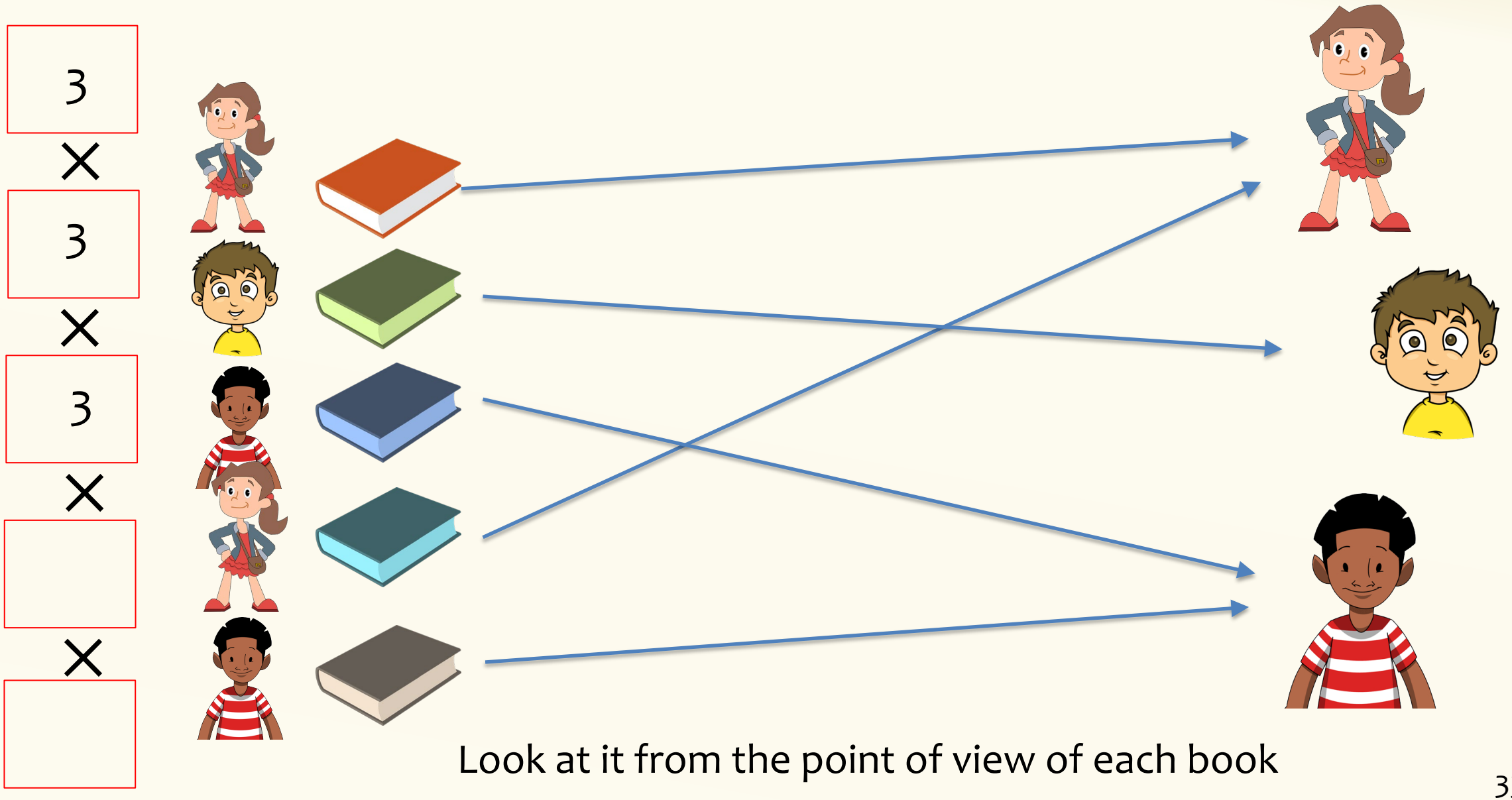
Look at it from the point of view of each book

Book assignments – A Clever Approach



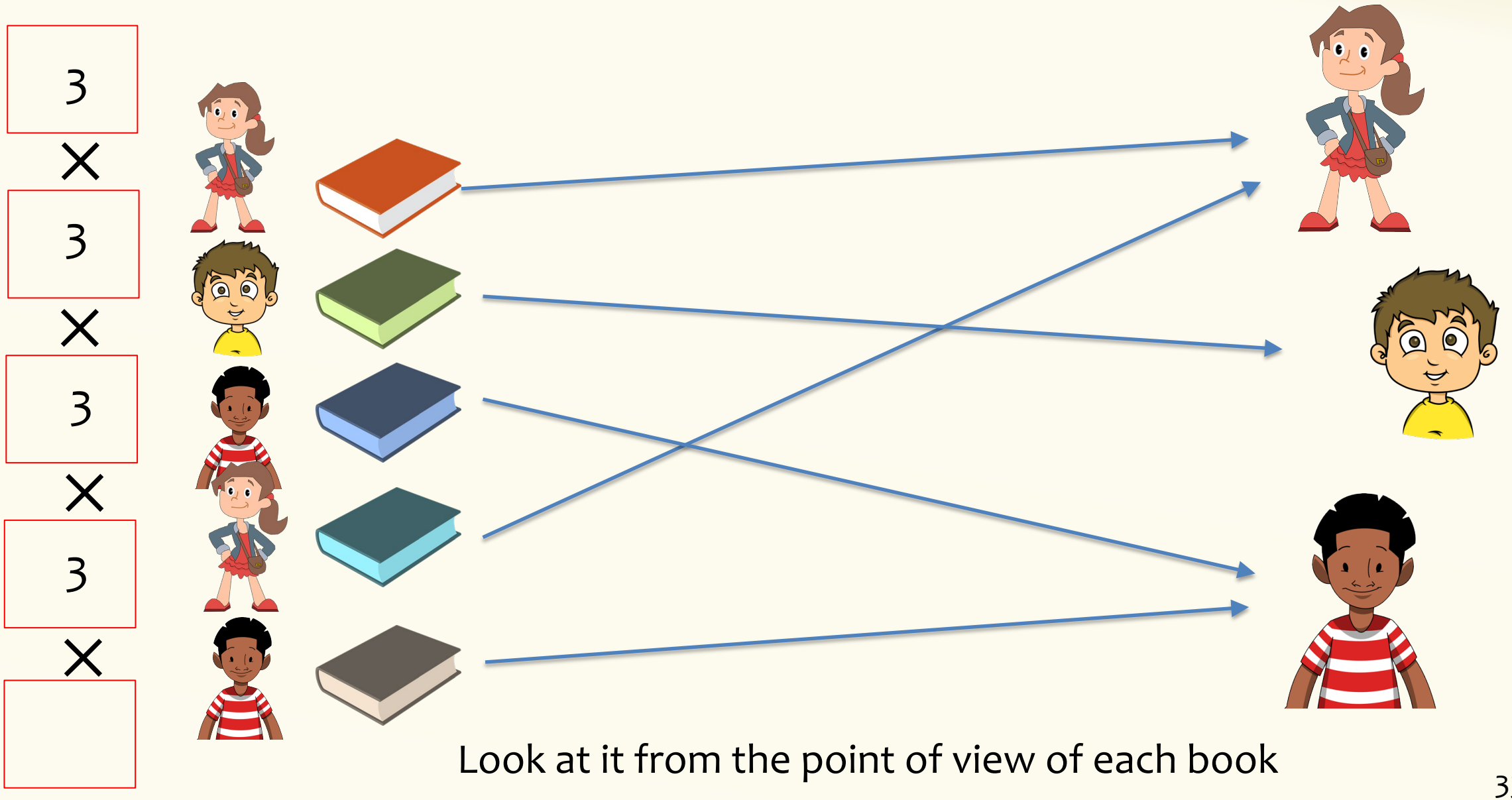
Look at it from the point of view of each book

Book assignments – A Clever Approach



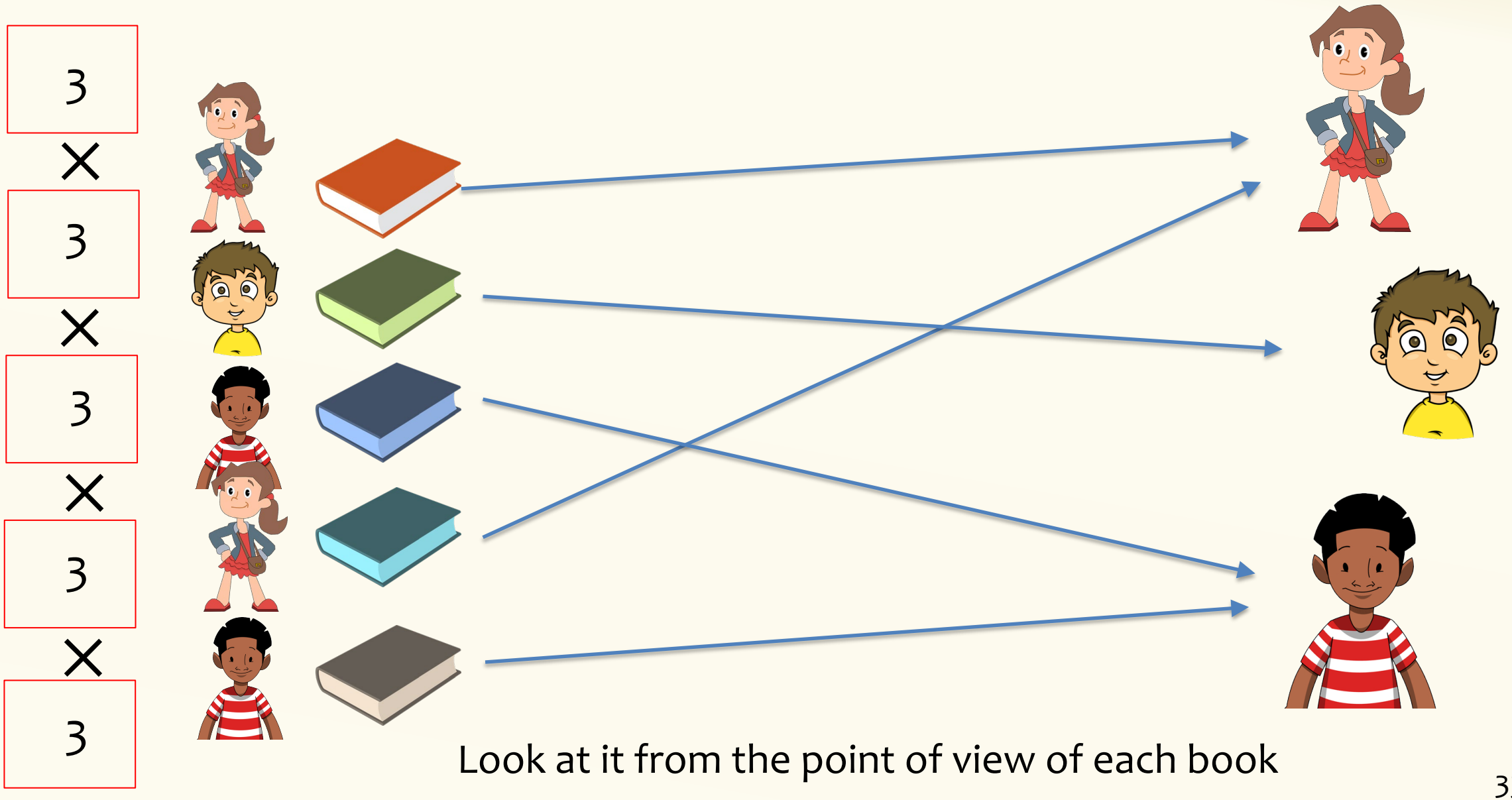
Look at it from the point of view of each book

Book assignments – A Clever Approach



Look at it from the point of view of each book

Book assignments – A Clever Approach



Look at it from the point of view of each book

Lesson: Representation of what we are counting is very important!

Tip: Use different methods to double check yourself
Think about counter examples to your own solution.

The first concept check is out and due at 1:00pm before Friday's lecture

The concept checks are meant to help you immediately reinforce what is learned.

Students from previous quarters have found them really useful!